

MARBLE POINT, AntarcticaWMO: **898660**Lat: **77.439S**Lon: **163.754E**Elev: **276**StdP: **14.55**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-34.8	-32.1	N/A	N/A	N/A	N/A	N/A	N/A	41.4	-3.7	34.5	-7.1	7.1	140	1.189

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	8.5	38.3	N/A	36.0	N/A	34.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.2	26.2	22.0	DB	-39.9	43.6	5.3	2.9	-43.7	45.7	-46.8	47.4	-49.8	49.1	-53.7	51.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	27.9	18.3	2.9	-6.1	-7.3	-9.9	-13.6	-14.2	-9.6	2.0	18.3	28.1		
	DBStd	17.70	4.43	6.55	7.73	9.89	11.07	10.23	11.24	11.47	10.05	8.05	7.10	4.50		
	HDD50	17160	685	888	1459	1683	1777	1798	1970	1991	1789	1489	952	679		
	HDD65	22635	1150	1308	1924	2133	2242	2248	2435	2456	2239	1954	1402	1144		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	8.5	7.8	9.0	8.2	7.3	8.3	9.5	8.7	8.8	9.3	8.2	8.7	8.2		
	PrecAvg	8.3	0.9	0.9	0.7	0.5	0.6	0.7	0.5	0.7	0.7	0.7	0.7	0.7		
(15) Precipitation	PrecMax	18.5	1.9	2.4	2.7	1.4	1.9	2.4	1.5	2.9	2.3	2.4	2.0	2.4		
	PrecMin	1.9	0.1	0.0	0.0	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	4.3	0.6	0.7	0.6	0.3	0.5	0.6	0.4	0.7	0.5	0.6	0.4	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	34.8	22.3	20.9	22.4	20.0	18.6	18.1	17.8	23.4	35.4	42.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	38.3	31.7	19.0	15.8	16.5	13.7	12.4	11.8	13.4	18.8	32.2	38.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	36.1	29.2	16.0	11.9	12.3	9.0	7.4	7.7	9.5	15.7	29.8	36.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	34.1	27.1	13.4	7.4	8.1	4.9	2.7	3.8	5.0	12.9	27.5	34.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.5	9.1	9.9	10.8	11.3	11.6	12.8	12.8	12.3	10.5	8.5		
		MCDBR	10.3	10.9	11.7	12.2	13.8	15.1	14.5	14.9	15.7	14.8	12.0	11.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.220	0.176	0.112	N/A	N/A	N/A	0.110	0.165	0.196	0.216	0.232		
	taud		2.324	2.336	2.227	1.911	N/A	N/A	N/A	1.894	2.169	2.290	2.322	2.322		
	Ebn at Noon		310	288	248	51	0	0	0	44	247	297	316	317		
	Edh at Noon		29	23	14	1	0	0	0	1	15	23	29	31		
(44) All-Sky Solar Radiation	RadAvg		2573	1519	534	48	0	0	0	8	289	1168	2372	2969		
	RadStd		72	50	21	2	0	0	0	1	15	37	62	81		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)		+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page